Name: _____

TOPIC TEST 2

Chapter 3: Arithmetic and geometric sequences

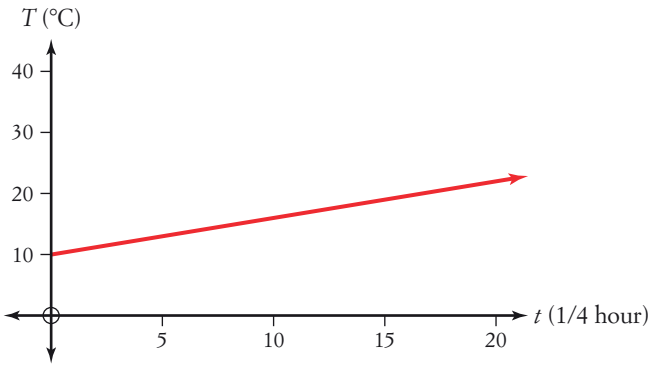
- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 9 short-answer questions (28 marks)
- Part C: 4 analysis questions (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 At the beginning of a particular day the temperature was 10°C. It became steadily warmer during the day, increasing 0.6°C every quarter of an hour. The temperatures over a period of 5 hours are shown below.



After how many hours, to the nearest hour, is the temperature 20°C?

A 2

B 4

C 5

D 17

E 16

[1 mark]

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2 The first four terms of a sequence are 2, 8, 14, 20, 26. The 10th term of the sequence is:

A 2

B 32

C 56

D 50

E 290

[1 mark]

3 The first five terms of a sequence are 2, 8, 14, 20, 26. The sum to the 10th term of the sequence is:

A 320

B 32

C 56

D 50

E 290

[1 mark]

4 The common difference $d = t_n - t_{n-1}$ for the sequence 1, −8, −17, −26, −35, … is:

A 9

B −9

C 10

D −10

E −44

[1 mark]

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5 The common ratio $r = \frac{t_n}{t_{n-1}}$ for the sequence 1, −4, 16, −64, … is:

A 9

B −9

C 4

D −4

E 256

[1 mark]

Part B

Short-answer questions

9 short-answer questions
28 marks
Show your working where appropriate.

1 Find the arithmetic sequence if the second term in the sequence is 3 and the ninth term is −10.

[4 marks]

2 If the first six terms of a sequence are 2, 2.5, 3, 3.5, 4, 4.5, state the sum of the sequence to 10 terms.

[3 marks]

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3 Find the sum of the arithmetic sequence 30, 27, 24, 21, …, −111.

[4 marks]

4 Calculate the sum of all the multiples of 6 between 1 and 100.

[4 marks]

5 If x, $x + 3$, and $x + 5$ are the first three terms of a geometric sequence, calculate the value of x.

[2 marks]

6 Find the first four terms of the sequence described by the recursion:

$$a_1 = 6 \text{ and } a_n = -3a_{n-1}.$$

[2 marks]

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7 If the first five terms of a sequence are $-2, 6, -18, -243, 729, -2187$, state the value of the 17th term of the sequence.

[3 marks]

8 Find the common ratio $r = \frac{t_n}{t_{n-1}}$ for the sequence $1, -\frac{1}{3}, \frac{1}{9}, -\frac{1}{27}, \frac{1}{81}, \ldots$

[2 marks]

9 Find the geometric sequence if the second term in that sequence is 3 and the ninth term is 354.

[4 marks]

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Part C

Analysis questions

4 analysis questions
18 marks
Show your working where appropriate.

1 James takes out a personal loan of $8500 to buy a second-hand car, and is charged 9% every year in interest on the remaining balance. James is unable to pay anything off the balance of his loan.

a Tabulate the balance of James' loan over a period of 11 years.

b After how many years does James owe $20 000?

[4 marks]

2 Find a value(s) of x such that $\frac{1}{x^2}, \frac{x}{4}, \frac{x^3}{6}, \ldots$ will form successive terms of a geometric sequence.

[4 marks]

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3 Curtains are installed in layers in a home. One layer lets in 95% of light.

a What percentage of light is let in by:

i two layers

ii three layers

iii 10 layers of curtaining?

b How many layers will let in 30% of light?

[5 marks]

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4 A ball is dropped from a height of 1 metre and bounces up to $\frac{2}{3}$ of its original height. It continues bouncing, rising $\frac{2}{3}$ of its height on each bounce until it eventually reaches the ground.

a What is the total distance through which it travels?

b Will the ball ever stop bouncing?

[5 marks]

[Total marks: 51]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $a = 10, d = 0.6$, so $t_n = 10 + (n - 1)(0.6)$

$20 = 10 - 0.6 + 0.6n$

$n = \dfrac{10.6}{0.6} = 17.7$ quarter hours

$\dfrac{17.7}{4} = 4.4$ hrs (round to 4)

$\therefore$ **B**

2 $a = 2$

$d = 8 - 2 = 6$

$t_n = 2 + 6(n - 1)$

$t_{10} = 2 + 6(10 - 1)$

$= 2 + 6 \times 9$

$= 56$

$\therefore$ **C**

3 $a = 2$

$d = 8 - 2 = 6$

Use the formula:

$S_n = \dfrac{n}{2}[2a + (n - 1)d]$

$S_{10} = \dfrac{10}{2}[4 + 9 \times 6]$

$= 5(4 + 54)$

$= 290$

$\therefore$ **E**

4 $d = t_n - t_{n-1}$

$= -8 - 1$

$= -17 - (-8)$

$= -26 - (-17)$

$= -35 - (-26)$

$= -9$

$\therefore$ **B**

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5 $r = \dfrac{t_n}{t_{n-1}}$

$= \dfrac{-4}{1} = \dfrac{16}{-4}$

$= -4$

$\therefore$ **D**

Part B – Short-answer questions

1 $t_2 = 3$ and $t_9 = -10$ **[1 mark]**

Use $t_n = a + (n-1)d$ to solve simultaneously for a and d.

$a + (2-1)d = 3$ and $a + (9-1)d = -10$ **[1 mark]**

gives $a = \dfrac{34}{7}, d = -\dfrac{13}{7}$ **[1 mark]**

The arithmetic sequence is $\dfrac{34}{7}, \dfrac{21}{7}, \dfrac{8}{7}, \ldots$ **[1 mark]**

2 For the sequence 2, 2.5, 3, 3.5, 4, 4.5, $a = 2$, $d = 0.5$. **[2 marks]**

Use $S_n = \dfrac{n}{2}[2a + (n-1)d]$ to find S_{10}.

$S_{10} = \dfrac{10}{2}[2(2) + (10-1)0.5]$ **[1 mark]**

$\therefore S_{10} = 42.5$

3 For the arithmetic sequence 30, 27, 24, 21, …, -111, $a = 30$, $d = -3$, $l = -111$. **[2 marks]**

Use $S_n = \dfrac{n}{2}(a + l)$ where $l = t_n = a + (n-1)d$.

Solve for n, $l = t_n = -111 = 30 + (n-1)(-3)$ gives $n = 48$. **[1 mark]**

$\therefore S_{48} = \dfrac{48}{2}(30 - 111)$ **[1 mark]**

$\therefore S_{48} = -1944$

4 Arithmetic sequence is 6, 12, 18, 24, …, 96, where $a = 6$, $d = 6$, $l = 96$. **[2 marks]**

We need to choose 96, because that is the last multiple of 6 before 100.

Use $S_n = \dfrac{n}{2}(a + l)$ where $l = t_n = a + (n-1)d$.

Solve for n, $l = t_n = 96 = 6 + (n-1)(6)$ gives $n = 16$. **[1 mark]**

$\therefore S_{16} = \dfrac{16}{2}(6 + 96)$ **[1 mark]**

$\therefore S_{16} = 816$

The other sum formula could also be used, using $n = 16$.

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5 Using $r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \frac{t_4}{t_3} = \ldots$ gives $r = \frac{x+5}{x+3} = \frac{x+3}{x}$. **[1 mark]**

$x(x+5) = (x+3)(x+3)$

$x^2 + 5x = x^2 + 6x + 9$

$0 = -x + 9$

$x = -9$ **[1 mark]**

6 Sequence is $6, -18, 54, -162$ **[2 marks]**

7 Sequence is $-2, 6, -18, -243, 729, -2187$, where $a = -2$, $r = -3$. **[2 marks]**

Using $t_n = ar^{n-1}$ to find t_{17}

$t_{17} = (-2)(-3)^{17-1}$ **[1 mark]**

$\therefore t_{17} = -86\,093\,442$

8 Using $r = \frac{t_n}{t_{n-1}}$ to find $r = \frac{-\frac{1}{3}}{1}$ **[1 mark]**

Gives $r = -\frac{1}{3}$. **[1 mark]**

9 $t_2 = 3$ and $t_9 = 354$ **[1 mark]**

$t_n = ar^{n-1}$ to solve simultaneously for a and r.

$ar^{2-1} = 3$ and $ar^{9-1} = 354$ **[1 mark]**

gives $a = \frac{3}{\sqrt[7]{118}}$, $r = \sqrt[7]{118}$ **[1 mark]**

The geometric sequence is $\frac{3}{\sqrt[7]{118}}, 3, 3\sqrt[7]{118}, \ldots$ **[1 mark]**

Part C – Analysis questions

1 a

Year	Loan balance
1	8500
2	9625
3	10 099
4	11 008
5	11 999
6	13 079
7	14 256
8	15 539
9	16 936
10	18 460
11	20 121

[2 marks]

b James owes $20 000 after 11 years. **[2 marks]**

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2 Using $r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \frac{t_4}{t_3} = \cdots$ gives $r = \frac{\frac{x^3}{6}}{\frac{x}{4}} = \frac{\frac{x}{4}}{\frac{1}{x^2}}$ **[2 marks]**

$$r = \frac{x^3}{6} \times \frac{4}{x} = \frac{x}{4} \times \frac{x^2}{1}$$

$$\frac{4x^3}{6x} = \frac{x^3}{4}$$

$$4 \times 4x^3 = x^3 \times 6x$$

$$16x^3 = 6x^4$$

$$8 = 3x$$

$$x = \frac{8}{3}$$ **[2 marks]**

3 a i If one layer of curtain lets in 95% of light we can say that:

$0.95^n = x$, where $n = 1$ curtain layer and $x =$ percentage of light that is let in.

If we have two curtain layers, then $n = 2$, so we get:

$0.95^2 = x$

$x = 0.9025$

$\therefore$ Two curtain layers let in 90.25% of light. **[1 mark]**

ii $0.95^3 = x$

$0.95^3 = 0.8574$

$\therefore$ Three curtain layers let in 85.74% of light. **[1 mark]**

iii $0.95^{10} = x$

$0.95^{10} = 0.5987$

$\therefore$ Ten curtain layers let in 59.87% of light. **[1 mark]**

b Solve $0.95^n = 0.3$

$\therefore n = 23$

Therefore, 23 layers will be needed to block out 30% of light. **[2 marks]**

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4 a Use $S_\infty = \dfrac{a}{1-r}$ with $a = \dfrac{2}{3}, r = \dfrac{2}{3}$. **[2 marks]**

Distance travelled is original 1 metre plus twice the height of each subsequent bounce.

Total distance travelled =

$$1 + 2\left(\frac{a}{1-r}\right)$$

$$= 1 + 2\left(\frac{\frac{2}{3}}{1-\frac{2}{3}}\right)$$ **[2 marks]**

$$= 1 + 2\left(\frac{\frac{2}{3}}{\frac{1}{3}}\right)$$

$$= 5$$

= 5 metres

b As this is a limiting sum, in theory, the ball will never stop bouncing. **[1 mark]**

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